

Trade-offs, Comparative Advantage, and the Market System

ECON 101 - Macroeconomics

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Fall 2026

- **2.1 Production Possibilities Frontiers and Opportunity Costs**
- 2.2 Comparative Advantage and Trade
- 2.3 The Market System

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Scarcity and Trade-offs

- People continually face decisions about how best to use their **scarce** resources.
- **Scarcity**: a situation in which unlimited wants exceed the limited resources available to fulfill those wants.
- Scarcity requires **trade-offs**: to get more of one thing, society must accept less of another.
- Economics gives us tools to help make these trade-offs wisely.

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- Tesla has scarce workers and machinery.
- If Tesla wants to produce more Model 3 sedans, those resources are not available to produce Cybertruck pickups (and vice versa).
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2.1 Learning Objective

Goal

Use a production possibilities frontier to analyze opportunity costs and trade-offs.

- A **production possibilities frontier (PPF)** is a curve showing the maximum attainable combinations of two goods that can be produced with available resources and current technology.
- The PPF is a **positive** tool: it shows what *is* possible, not what *should* be produced.

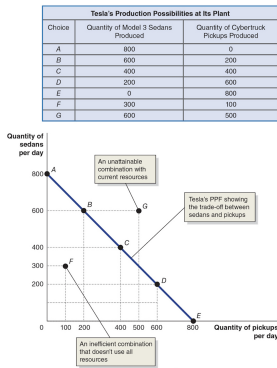
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Figure 2.1 – Tesla's Production Possibilities Frontier (1 of 2)



Interpreting the PPF: Tesla Example

- Tesla can produce combinations of Model 3 sedans and Cybertruck pickups.
- Points on the PPF: attainable and **productive efficient**.
- Points below the PPF: attainable but **inefficient** (resources underused).
- Points above the PPF: currently unattainable with existing resources and technology.

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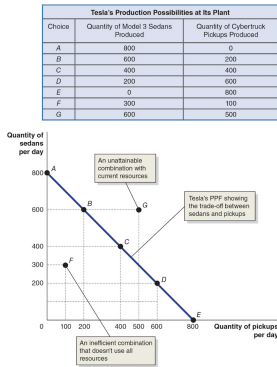
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Figure 2.1 – Tesla's PPF and Opportunity Cost (2 of 2)



Opportunity Cost on the PPF

- Suppose moving from point A to point B:
 - Tesla produces 200 more Cybertruck pickups.
 - To do so, it must produce 200 fewer Model 3 sedans.
- The **opportunity cost** of producing 200 more Cybertrucks is the 200 Model 3 sedans given up.
- **Opportunity cost**: the highest-valued alternative that must be given up to engage in an activity.

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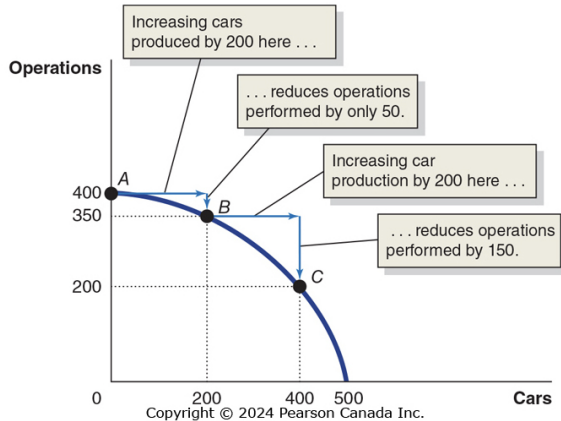
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Figure 2.2 – Increasing Marginal Opportunity Costs



Why Opportunity Costs Often Increase

- Some resources are better suited to one task than another.
- The first resources to be switched are the ones best suited to switching.
- As more resources are devoted to an activity, the extra output from additional resources tends to fall.
- This gives the PPF its bowed-out shape and leads to **increasing marginal opportunity cost**.

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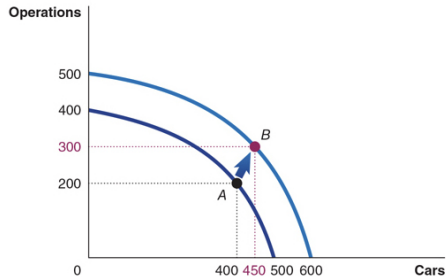
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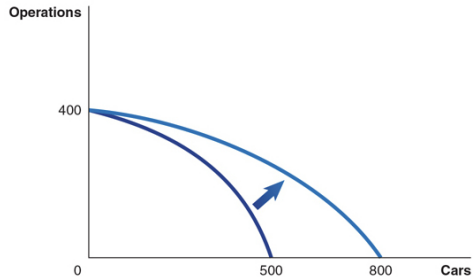
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Figure 2.3 – Economic Growth



(a) Shifting out the PPF



(b) Technological change in the automotive industry

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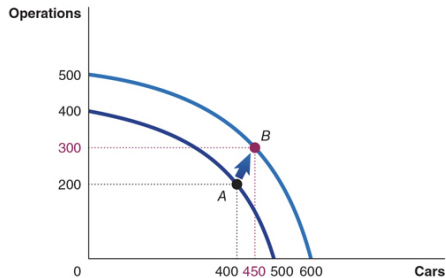
- As more economic resources become available, the economy can move from point A to point B, producing more of both goods.
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- Macro link: long-run growth allows higher living standards without giving up as much of other goods.

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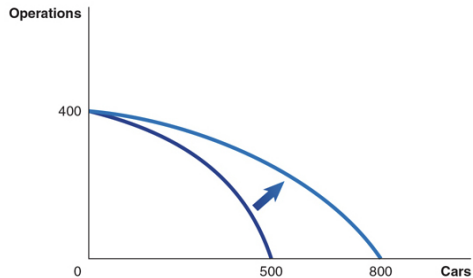
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Figure 2.3 – Technological Change in One Sector



(a) Shifting out the PPF



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Technological Change in One Industry

- Suppose technology improves in the automobile industry only.
- The maximum output of automobiles increases, but the production possibility of the other good (e.g., tanks) is unchanged.
- Previously unattainable combinations become attainable.
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2.2 Learning Objective

Goal

Understand comparative advantage and explain how it serves as the basis for trade.

Two-Person Production Example: You and Your Neighbour

- You and your neighbour each have limited time to pick apples and/or cherries.
- The table (in the slides) shows:
 - If you spend all your time picking cherries, you can pick 20 kg of cherries.
 - If you spend all your time picking apples, you can pick 20 kg of apples.
 - Your neighbour can pick 60 kg of cherries or 30 kg of apples if she specializes.
- We use these numbers to illustrate absolute and comparative advantage.

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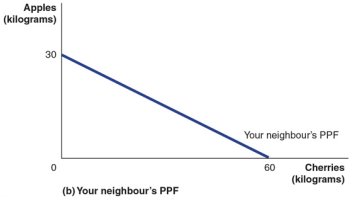
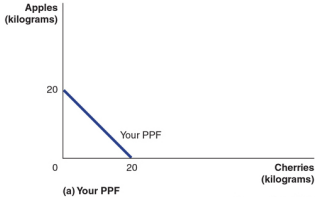
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Figure 2.4 – Production Possibilities for You and Your Neighbour, Without Trade

	You		Your Neighbour	
	Apples (kg)	Cherries (kg)	Apples (kg)	Cherries (kg)
Pick nothing but apples	20	0	30	0
Pick nothing but cherries	0	20	0	60



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Autarky (No Trade) Outcomes

- Without trade, each person consumes only what they produce.
- **You:** pick and consume 8 kg of apples and 12 kg of cherries per week (point A).
- **Neighbour:** picks and consumes 9 kg of apples and 42 kg of cherries per week (point C).
- Both are on their own PPFs but may not be at the best possible consumption points if trade is allowed.

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Specialization and Gains from Trade

- **Trade:** the act of buying and selling.
- What if you and your neighbour decide to specialize and trade?
- Your neighbour appears better at picking both apples and cherries, but:
 - You can still both benefit from trade by specializing in what you are **relatively** good at.

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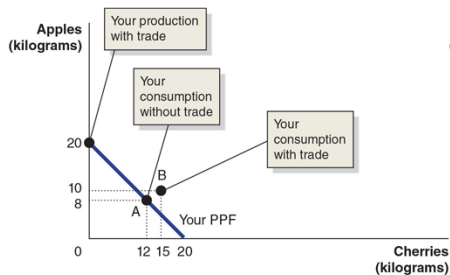
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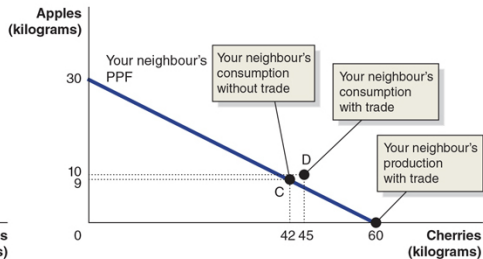
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Figure 2.5 – Gains from Trade (1 of 2)



(a) Your production and consumption with trade



(b) Your neighbour's production and consumption with trade

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Numerical Trade Example

- If you specialize in picking apples, you can pick 20 kg.
- If your neighbour specializes in picking cherries, she can pick 60 kg.
- Suppose you trade 10 kg of your apples for 15 kg of your neighbour's cherries:
 - You consume 10 kg of apples and 15 kg of cherries (point B).
 - Neighbour consumes 10 kg of apples and 45 kg of cherries (point D).
- Both are better off than in autarky.

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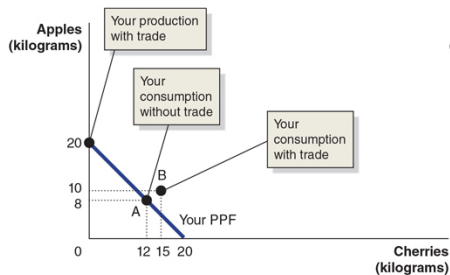
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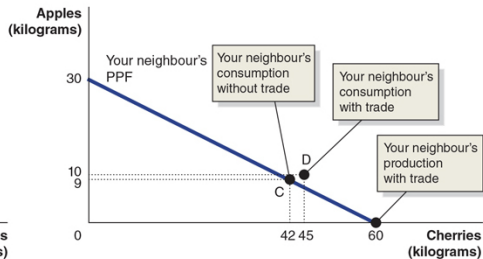
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Figure 2.5 – Gains from Trade (2 of 2)



(a) Your production and consumption with trade



(b) Your neighbour's production and consumption with trade

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Table 2.1 – Summary of the Gains From Trade

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	You		Your Neighbour	
	Apples (kg)	Cherries (kg)	Apples (kg)	Cherries (kg)
Production and consumption <i>without</i> trade	8	12	9	42
Production <i>with</i> trade	20	0	0	60
Consumption <i>with</i> trade	10	15	10	45
Increased consumption (gains from trade)	2	3	1	3

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Table 2.1 – Summary of the Gains From Trade

- Both you and your neighbour are able to consume more with trade than without trade.
- Key points to emphasize to students:
 - Trade allows consumption points outside each individual's own PPF.
 - Even if one person is better at producing *both* goods, both can gain.

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Explaining the Gains from Trade

- Your neighbour has an **absolute advantage** in both cherry and apple picking.
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	Opportunity Cost of Apples	Opportunity Cost of Cherries
You	1 kg of cherries	1 kg of apples
Your neighbour	2 kg of cherries	0.5 kg of apples

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Table 2.2 – Opportunity Costs of Picking Apples and Cherries

- This table compares the opportunity cost of apples vs. cherries for you and your neighbour.
- The key lesson:
 - The basis for trade is **comparative advantage**, not absolute advantage.
 - Individuals, firms, and countries are better off if they specialize in producing goods and services for which they have a comparative advantage and obtain the other goods and services they need by trading.

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2.3 Learning Objective

Goal

Explain the basic idea of how a market system works.

- A **market** is a group of buyers and sellers of a good or service, and the institution or arrangement by which they come together to trade.
- Two key groups participate in the modern economy:
 - Households
 - Firms

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- **Capital:** physical capital such as computers, machines, and buildings used to produce other goods.
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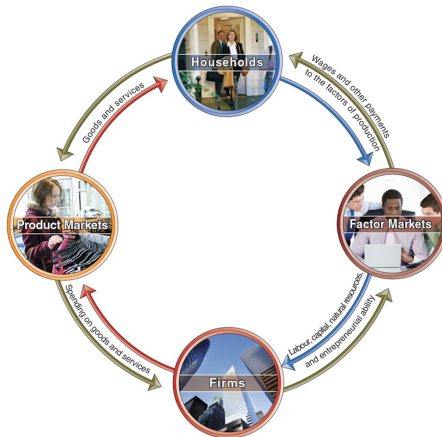
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Figure 2.6 – The Circular-Flow Diagram (1 of 2)



Circular-Flow Diagram: Basic Idea

- The **circular-flow diagram** is a model that illustrates how participants in markets are linked.
- Real flows:
 - Households provide factors of production to firms.
 - Firms provide goods and services to households.
- Money flows:
 - Firms pay money to households for the factors of production.
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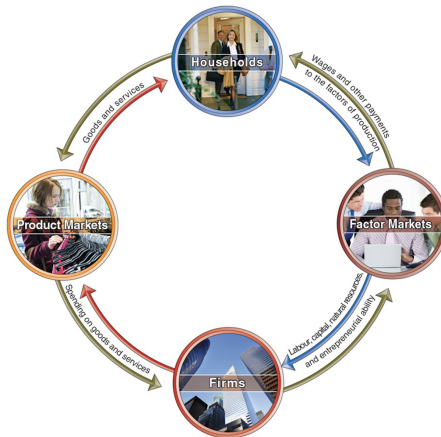
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Figure 2.6 – The Circular-Flow Diagram (2 of 2)



Limits of the Simple Circular-Flow Model

- Like all economic models, the circular-flow diagram is a **simplified** version of reality.
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- Countries that come closest to this benchmark have typically been more successful than centrally planned economies in raising living standards.
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Things to Remember

- The production possibilities frontier should **never bow inward**. Why?
 - Because opportunity cost typically increases as you move along the PPF.
- The PPF tells us what **can** be produced, not what **should** be produced.
- Just because someone is better or worse at *everything* does not mean trade with them cannot be beneficial.
- The basis for trade is **comparative advantage**, not absolute advantage.
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